K-2 About the FastBack® Model 260E-G2

The FastBack® Model 260E-G2 is a horizontal motion conveyor. Product moves through a cycle that has a slow forward, fast backward motion. The horizontal motion of the FastBack® causes the flow rate (the rate at which product travels along the conveyor) to be proportional to the motion frequency of the conveyor. A faster flow rate results in lower bed depth (the height of the product in the pan).

The FastBack® Model 260E-G2 comes from the factory set up to run at its optimal speed of 60 Hz. This speed produces the fastest flow rate with the least mechanical stress. Operating the FastBack® between 60 and 65 Hz may improve travel rates for certain products, but should be done only after consulting a Heat and Control representative. Operating the FastBack® at more than 65 Hz will not improve flow rates.

WARNING

Operating the FastBack® at more than 65 Hz and/or contrary to prescribed VFD settings (see TechPub PH009) may contribute to early failure of key mechanical components and may void the product warranty.

The FastBack® Model 260E-G2 consists of two major subassemblies: the pan assembly and the drive assembly (see Photo 2-1). The sections that follow describe these sub-assemblies in detail.

Photo 2-1 – Pan and Drive Subassemblies

Pan Assembly

The pan assembly attaches to the top of the drive assembly with four 1/2-inch bolts found at the top of each support arm. Many standard pan accessories are available: the adjustable nose discharge, fines screen section, flow levelers, End-gates, Mid-gates, and Revolution gates. Typical pan finishes are smooth bead blast and rigidized. (Note: special pan finishes such as mirror finish and UHMW inserts are available on an application-specific basis.)

Adjustable Nose Discharge

The adjustable nose discharge (see Photo 2-2) guides product to a weigher or scale. It allows the pan length to be adjusted for precise positioning of the product. The adjustable nose discharge attaches to the pan with two four-prong knobs.

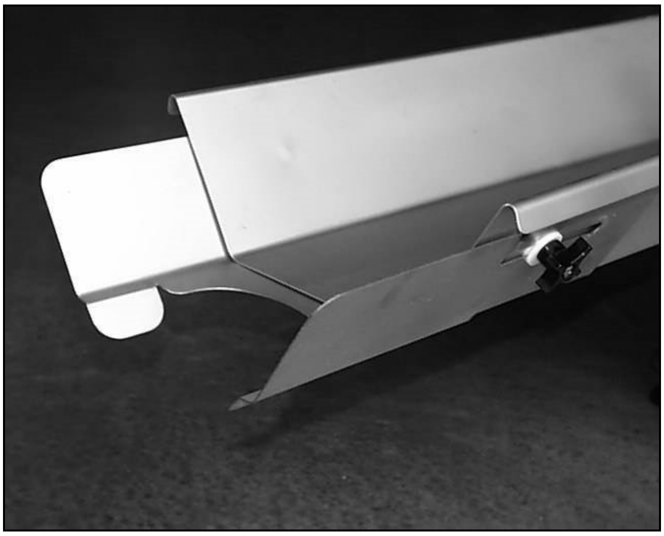

Photo 2-2 – Adjustable Nose Discharge

Flow Levelers

Flow levelers (see Photo 2-3) are used to achieve consistent bed depth from front to rear and side to side in applications that require greater process control. They are typically used on modulating pans, weigher feeder pans, and prior to seasoning drums. They can be either fastened to the pan or welded to the pan. Additionally, they are available in either large or small sizes.

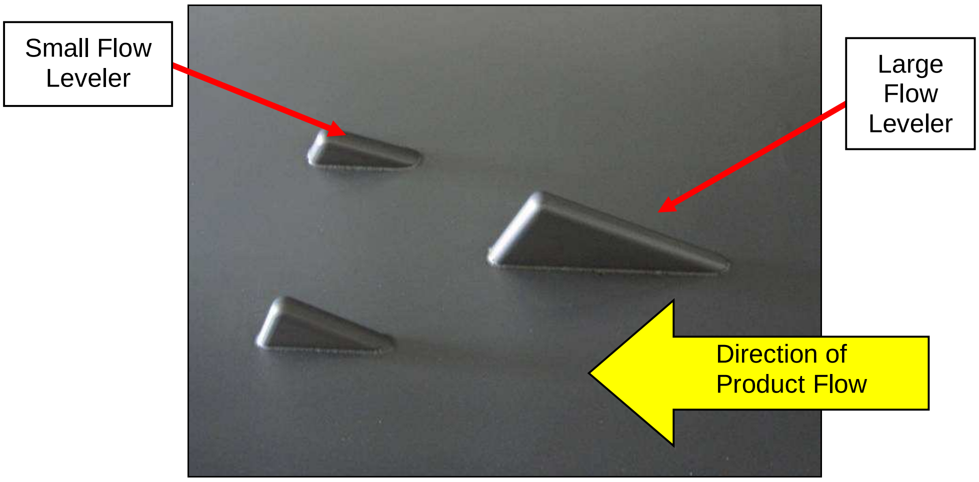

Photo 2-3 – Flow Levelers

Fines Screen Insert

The fines screen insert (see Photo 2-4 and Photo 2-5) is a drop-in accessory that removes product fines. It can be custom-engineered for specific product types. Blank inserts are also available when screening is not needed.

Photo 2-4 – Fines Screen Insert with Longitudinal Slits

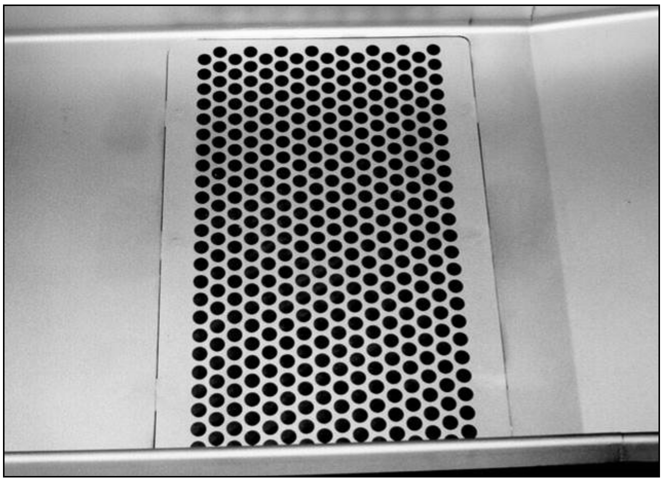

Photo 2-5 – Fines Screen Insert with Perforations

End-gates, Mid-gates, and Revolution Gates

There are three primary types of gates: End-gates, Mid-gates, and Revolution gates. End-gates are found at the discharge end of a pan (see Photo 2-6) or can be nested in the middle of a pan (see Photo 2-7).

Photo 2-6 – Example of an End-gate

Photo 2-7 – Example of a Nested End-gate

Mid-gates can be found at any point along the pan (see Photo 2-8 through Photo 2-10).

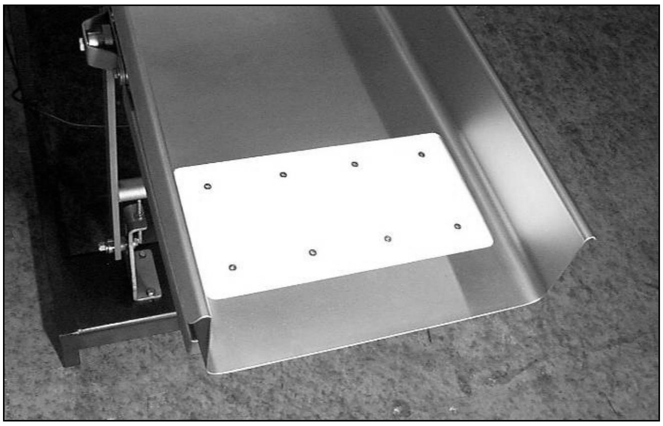

Photo 2-8 – Example of a Traditional Mid-gate

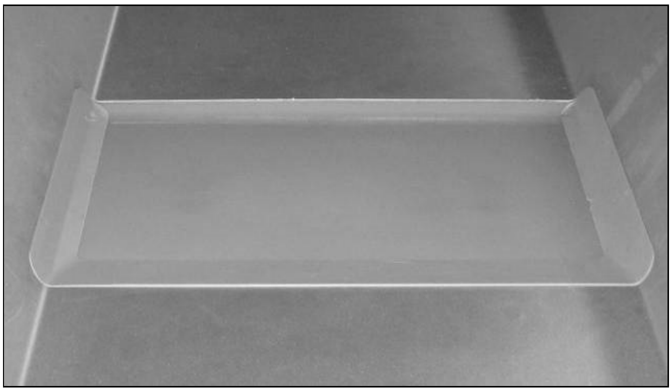

Photo 2-9 – Example of a Molded Polyurethane Mid-gate

Photo 2-10 – Example of a Mid-Revolution Gate

A Revolution gate, which is driven via an electric motor, is mounted onto the discharge end of a round-bottom pan and provides proportional feed rate capability (see Photo 2-11).

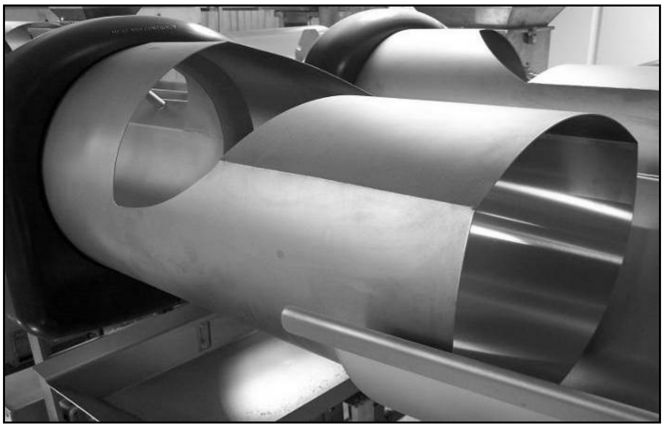

Photo 2-11 – Example of a Revolution gate

All sliding gates are controlled using a pressurized air system. This system can easily be disconnected at bracketed disconnect points (see Photo 2-12).

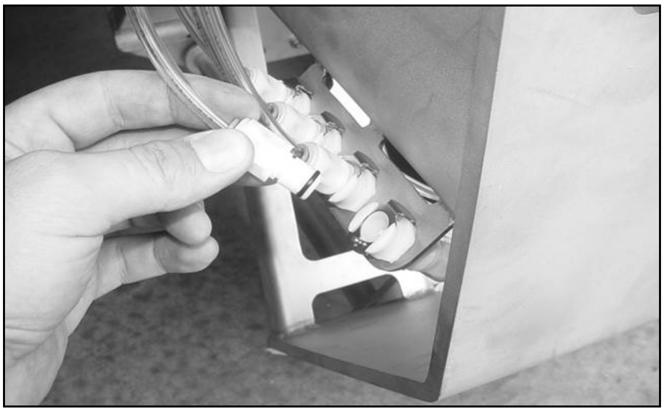

Photo 2-12 – Gate control disconnect bracket

An optional sliding gate accessory is the application of proximity sensors, which confirm that the gate is in either an open or closed position (see Photo 2-13).

Photo 2-13 – Example of End-gate with Position Proximity Sensors

Pan Finishes

Two typical pan finishes are offered and vary in use according to the FastBack® application. The most common finish is a smooth bead blasted finish (see Photo 2-14). The second most frequently used finish is a rigidized surface (see Photo 2-15), which is typically used for frozen products.

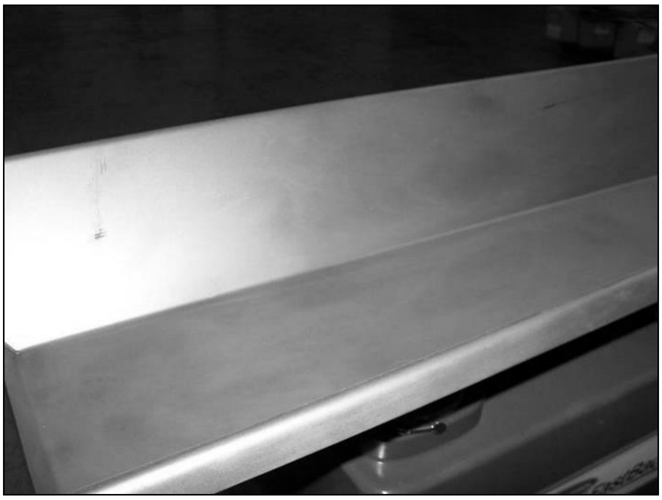

Photo 2-14 – Example of a smooth bead blasted finish

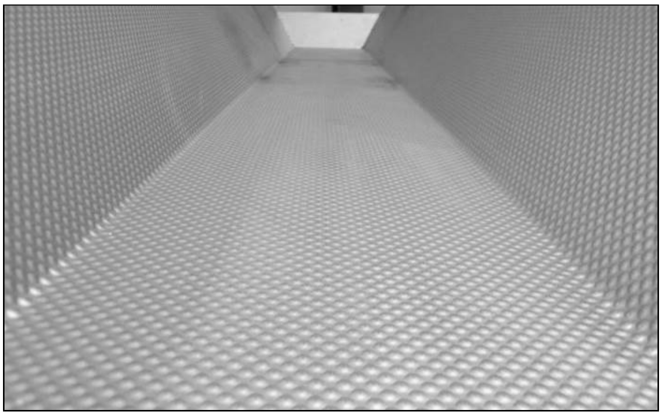

Photo 2-15 – Example of a rigidized finish

Drive Assembly

This section describes the various components that make up the drive assembly. These components are as follows:

- Enclosure
  - Utility Enclosure
  - Hand-held Operator
- Photoeye Assembly
- Level Transmitter
- Drive Unit Assembly
  - Motor Assembly
  - Transmission
    - Crankshaft Assembly
    - Drive Shaft
    - Tensioner Sprocket Assembly
    - Idler Sprocket Assembly
- Counterweight
- Articulating Feet

Enclosure

The four-piece enclosure covers the FastBack® drive assembly (see Photo 2-16). The primary enclosure piece is attached to the chassis with five (5) fasteners along the bottom perimeter of the chassis. The FastBack® 260E-G2 requires access only from one side to perform most maintenance functions.

CAUTION

Always remove the enclosure side panel before performing any maintenance on the drive assembly. Section 5 – Maintenance describes how to remove the enclosure side panel.

Photo 2-16 – Enclosure

Utility Enclosure

The FastBack® 260E-G2 utility enclosure houses the electrical and pneumatic components for the FastBack® conveyor. The enclosure is a NEMA 4X (IP65) stainless steel box that is attached to the outside of the drive assembly enclosure in-between the drive arms (see Photo 2-17).

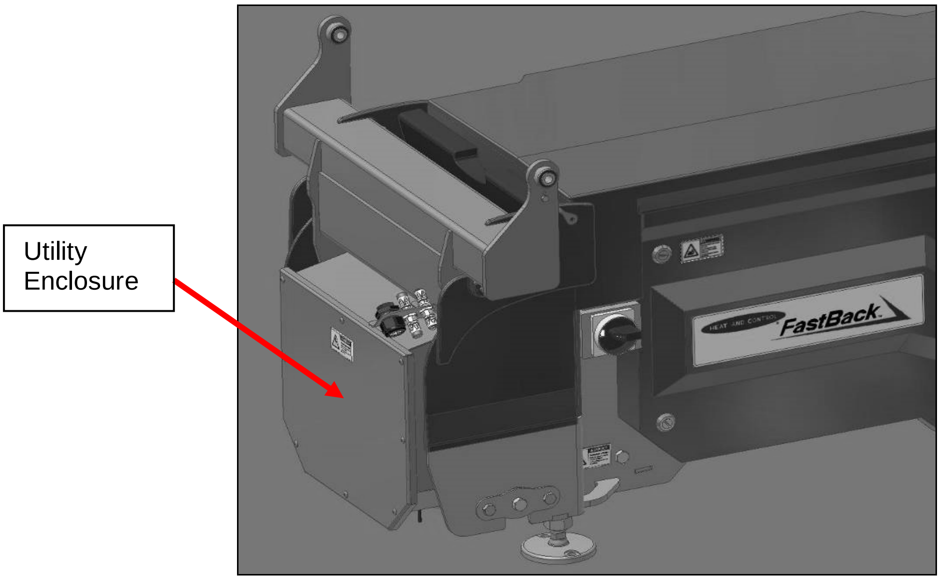

Photo 2-17 – Utility Enclosure

Based on the application, the utility enclosure may contain any of the following components:

- Up to two (2) VFDs (one for the FastBack® drive and one for the Revolution unit)
- Up to two (2) Line Reactors (<u>480 V</u> applications <u>ONLY</u> when the VFD is remotely mounted)
- Pneumatic pressure regulator and solenoid valves
- Terminal strips for control wiring

The utility enclosure comes from the factory wired to the drive assembly and initialized to run single speed, stop/start with run confirmation. At installation, the main power supply is wired into the disconnect switch(es) and the control input/output wires are landed on the terminal strip. Section 4 – Installation Procedures describes this process in more detail.

The default VFD for the FastBack® Model 260E-G2 is a 3 HP Yaskawa Model V1000. An alternate approved VFD is a 3 HP Allen-Bradley Model PowerFlex 70.

Hand-held Operator

A single hand-held operator with an I/O cable (see Photo 2-18) is supplied with each project. This hand-held unit allows access to the VFD parameters, error codes, and VFD status by connecting to a communication port located below the disconnect switch(es) (see Photo 2-19). See **Technical Publication PH009** or **Technical Publication PH015** for further details on VFD setup and parameter values.

Note: The hand-held operator is supplied only with your facility's original order. Spares are available through Heat and Control (see Appendix B).

Photo 2-18 – Yaskawa V1000 Hand-Held Operator

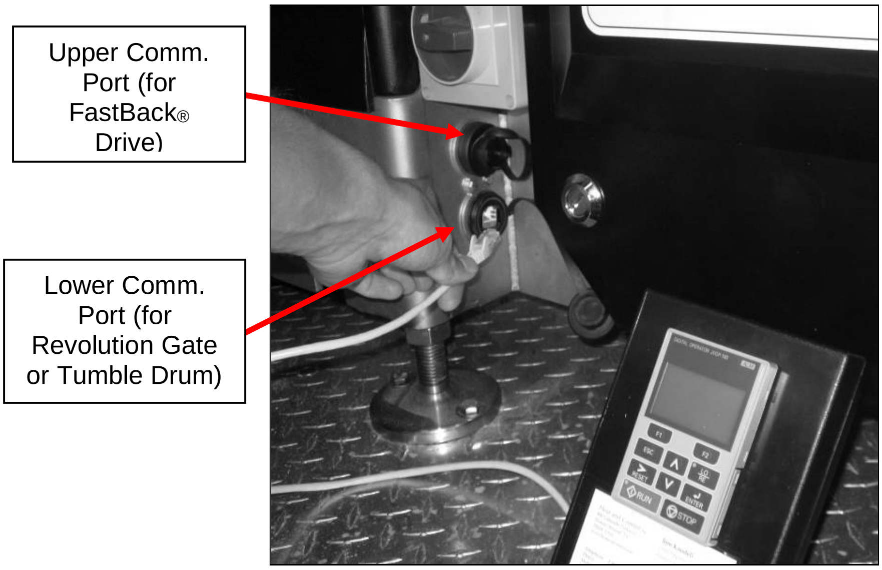

Photo 2-19 – Communication Port Access for Yaskawa V1000 Hand-Held Operator

Photoeye Assembly

The Photoeye assembly is a retro-reflective optical sensing system used to regulate the bed depth of product in the pan. It consists of an efector (or comparable) Photoeye, a quick-disconnect cable, and a mounting bracket (see Photo 2-20 and Photo 2-21). Most applications require only a single Photoeye. Two Photoeyes are supplied if two bed depth signals are desired.

Photo 2-20 – Photoeye Assembly with efector OGT500 Sensor

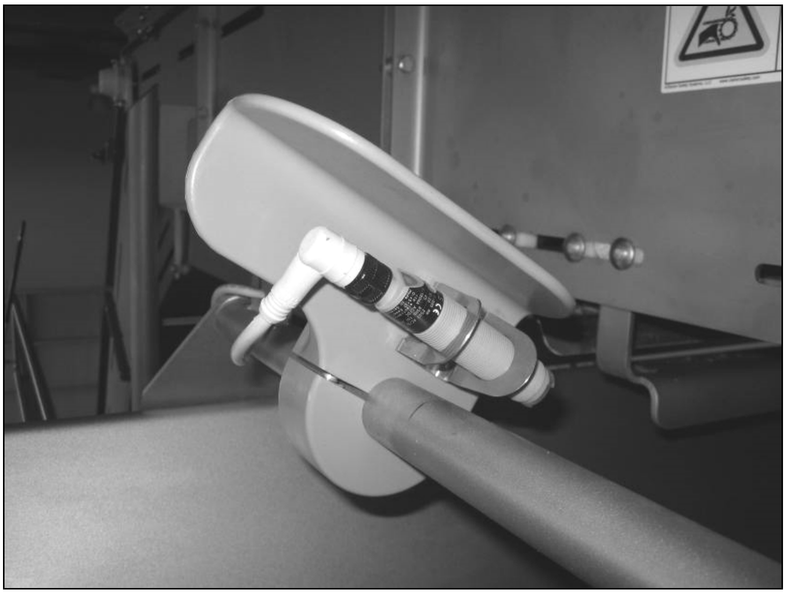

Photo 2-21 – Photoeye Assembly with efector OG35 Sensor

The 24V DC efector Photoeye is a diffused-reflective Photoeye (it returns the signal off the product itself) (see Figure 2-1). It has a maximum focal length of 600 mm (24 inches) and is suitable for a washdown environment.

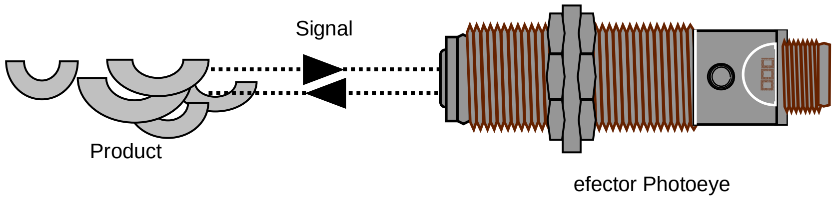

Figure 2-1 – efector Diffuse Photoeye

A junction box on the drive assembly enclosure connects the Photoeye assembly to the control source at your facility. During installation, the Photoeye must be wired to its power supply and to the control source. In addition, the brackets holding the Photoeye must be adjusted and tightened so that the Photoeye is properly positioned for calibration to the desired bed depth. See **Technical Publication PH006** for further details.

Level Transmitter

A level transmitter may be used to monitor changing bed depths and product travel rates. The ultrasonic sensor measures the distance between the height of the product in the pan and itself and returns a 4-20mA signal that is proportional to that distance. This signal may be used to control other equipment, such as a seasoning system. See **Technical Publication PH005** for further details.

Drive Unit Assembly

The remaining subassemblies described in this section reside within the enclosure. Photo 2-22 shows the drive unit with the enclosure side panel removed.

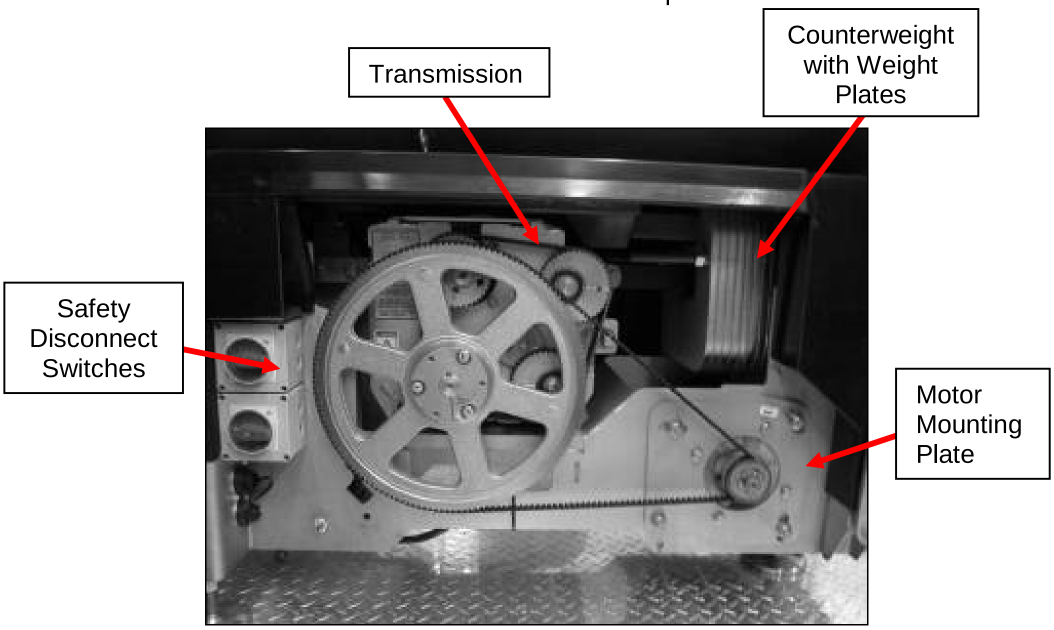

Photo 2-22 – Drive Enclosure Open

Motor Assembly

The motor is custom-engineered for the electro-mechanical load and space requirements of the 260E-G2 FastBack® (see Photo 2-23). The assembly consists of a totally enclosed, fan-cooled (TEFC) motor that is C-face mounted to an adjustable motor mounting plate and a drive sprocket mounted to the shaft.

Photo 2-23 – Motor (without sprocket or mounting plate)

Transmission

The transmission is one of three major sub-assemblies in the machine and is removable for rapid replacement when production time counts. The system consists of a removable transmission possessing four sprockets that work together to create the slow forward and fast backward motion of the FastBack® pan (see Photo 2-24).

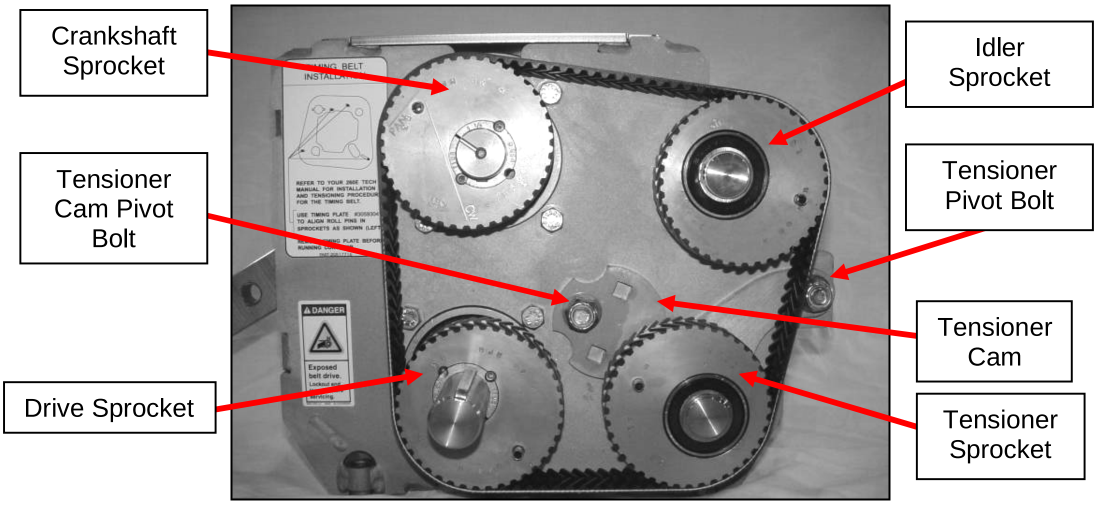

Photo 2-24 – Eccentric Sprocket System

- The drive sprocket and crankshaft sprocket are connected to the driveshaft and crankshaft, respectively.
- The tensioner sprocket is part of the tensioner sprocket assembly (see "Tensioner Sprocket Assembly" later in this section).
- The idler sprocket is part of the idler sprocket assembly (see "Idler Sprocket Assembly" later in this section).
- The tensioning cam applies the load required to tension the eccentric timing belt. Section 5 – Maintenance describes this procedure.

CAUTION

Descriptions of transmission sub-components are for informational purposes only. These sub-components are NOT sold individually. Heat and Control strongly recommends replacing worn transmissions with new or rebuilt whole assemblies from Heat and Control. Do not replace individual components of this assembly.

Note: ***Worn transmissions may be returned to the Heat and Control Galesburg office for guaranteed rebuild services. Contact the Galesburg office at (309) 342-5518.***

Crankshaft Assembly

The crankshaft assembly drives the pan and the counterweight. It consists of a crankshaft and two connecting rods, one for the drive arm and one for the counterweight (see Photo 2-25).

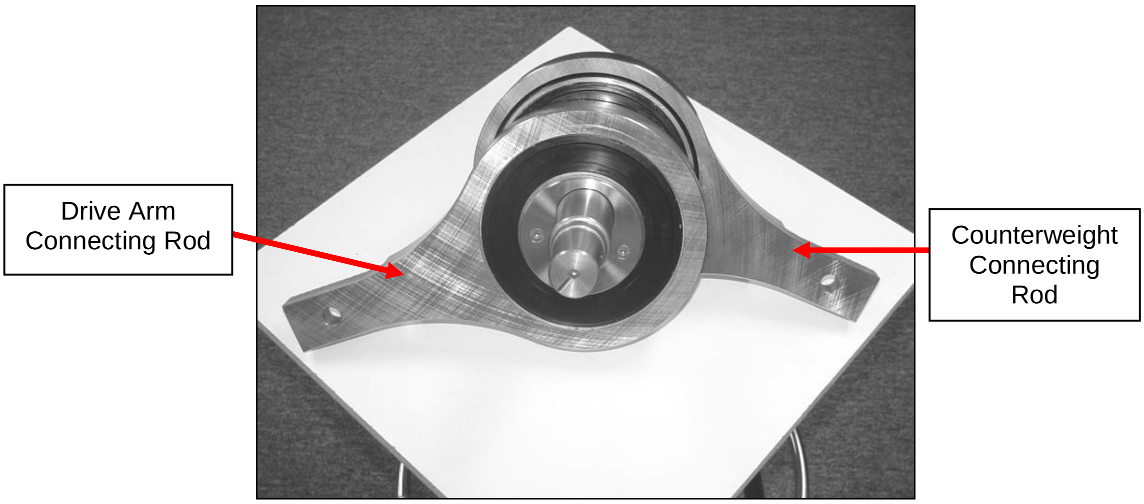

Photo 2-25 – Crankshaft Assembly

Drive Shaft

The drive shaft (see Photo 2-26) transfers power from the motor sprocket to the eccentric sprocket system. It consists of a polished OD shaft with a 1/4-inch key slot on one end.

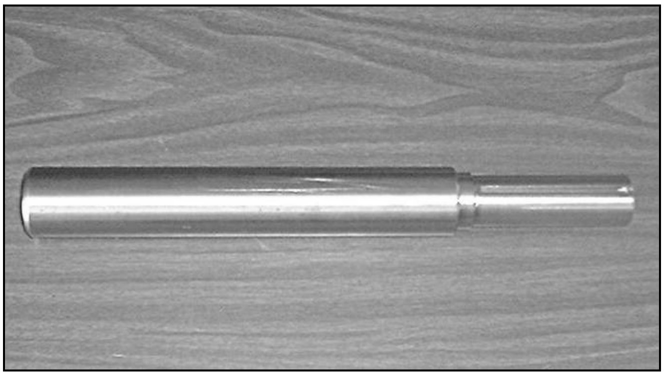

Photo 2-26 – Drive Shaft

Tensioner Sprocket Assembly

The tensioner sprocket assembly is part of the eccentric sprocket system and is used to tension the eccentric sprocket belt. It consists of an idler sprocket with two bearings mounted on a shaft. This shaft is welded to the tensioner arm and pivots about the narrow end. Photo 2-27 shows the tensioner sprocket assembly.

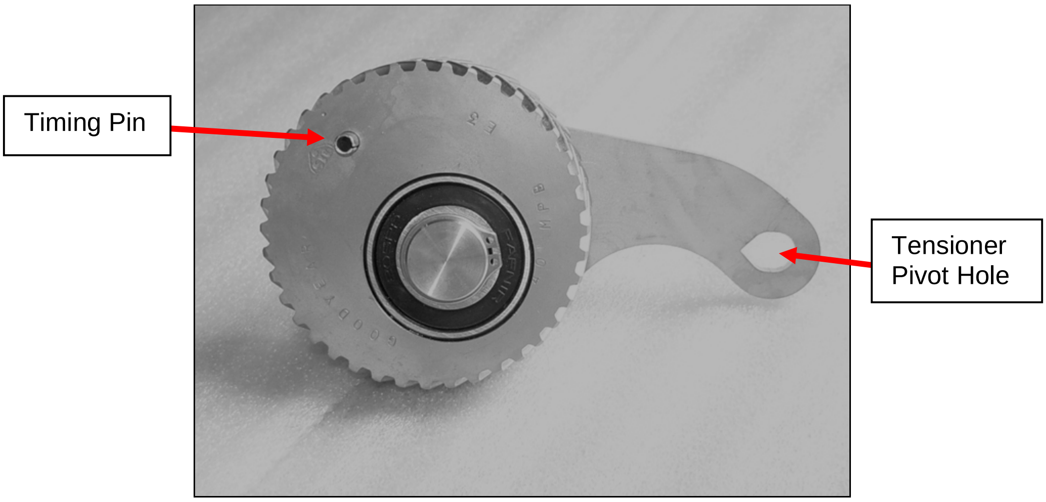

Photo 2-27 – Tensioner Sprocket Assembly

Idler Sprocket Assembly

The idler sprocket assembly takes up the slack in the eccentric timing belt during operation. It consists of a stationary idler in the eccentric sprocket system and an idler sprocket with two bearings mounted on a shoulder bolt. Photo 2-28 shows the idler sprocket.

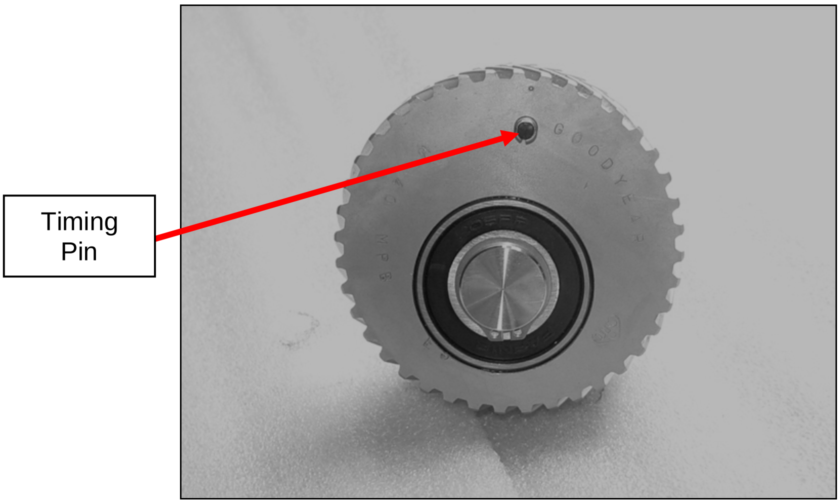

Photo 2-28 – Idler Sprocket

Counterweight

The counterweight compensates for the dynamic forces of the pan. It consists of a base billet and a stack of counterweight plates that are bolted to the billet. Each counterweight assembly is custom-engineered for a specific pan. Photo 2-29 shows a typical counterweight and its position inside the drive assembly.

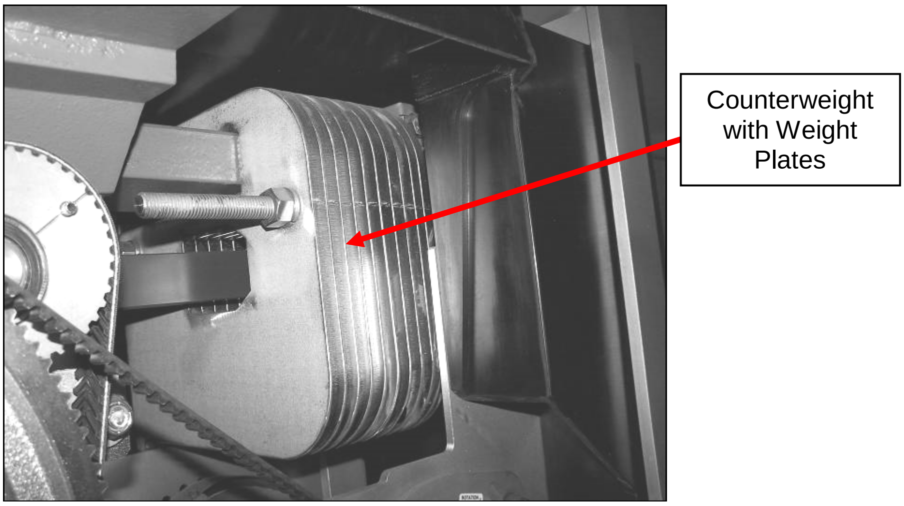

Photo 2-29 – Counterweight

Articulating Feet

The three articulating feet allow the FastBack® pan assembly to be positioned exactly as specified in the equipment layout drawing. The feet also enable the lock-down of the drive assembly to the support structure. Each articulating foot extends vertically +1.5 inches [38 mm] and -1.0 inches [25 mm], with 10° of swivel (see Photo 2-30).

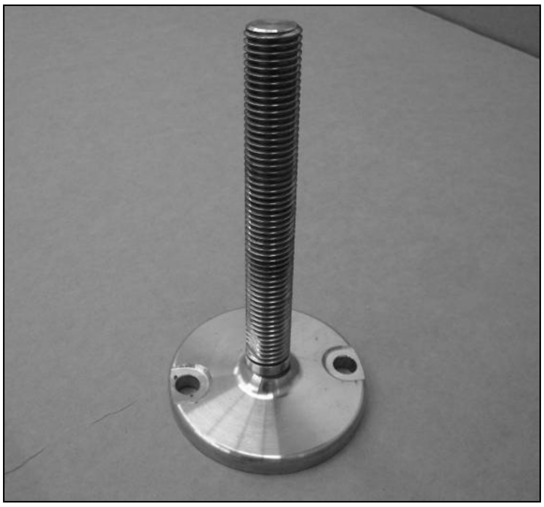

Photo 2-30 – Articulating Foot